Takeoff

PRESENTER GUIDE · 25 MINUTES · MURRAY STATE UNIVERSITY

Assume the room knows nothing. Not what a benchmark is, not what any of these models are called, not that any of this is measured at all. The demo is built for that audience and every screen explains itself — your job is mostly to stay out of the way and let people draw.

The bet: the room will draw five lines, get four of them badly wrong in the same direction, and then get the fifth wrong in the opposite direction. Everything after that explains why.

Before you start

- Open `index.html` in any browser. One file, no install, no network. If the venue wifi is broken it does not matter.
- Settings (⚙) → **Presenter mode** for a projector. Do it before the room fills.
- **Get people to actually draw.** The demo dies if you drive it and narrate. Hand the laptop round, or put the file on a screen and have people follow on their phones — it is built phone-first for exactly this.
- Each test has a **"see an example"** panel. Open it. For an audience new to this, one concrete task is worth more than any number you will say.
- Short on time: tests 1, 3 and 5. Test 5 is not optional — it is the point.

The opening screen (1 minute)

Read the first line aloud and then stop: *"You are going to draw five lines, and get four of them wrong."* The screen explains that AI systems are graded on public, checkable tests, and that this demo shows five of them. Do not add to it. Press the button.

Act I — the five tests (14 minutes)

> **Test 1 · Fixing real bugs · a third → four in five**
> Open the example: a real bug report, 1.2 million lines of code, write the patch, graded by the project's own tests. Ask for a number out loud before revealing — a spoken guess lands harder than a private one. Most rooms say 60–65%.
> **The line that matters comes after the number:** the labs stopped reporting this test in 2026. It was retired for being too easy, about two years after it was built to be hard.

> **Test 2 · Questions that beat PhDs · the two dashed lines**
> Point at the lines before anyone draws. 69.7% is a PhD answering in their own subfield. 34% is a clever non-expert with Google and half an hour. Both measured, both from the test's own paper.
> *"The PhD line was crossed in September 2024. Nobody threw a party. It was a Thursday."*

> **Test 3 · How long it can work alone · the one people misread**
> **Say this before they draw, every single time:** the scale is how long the job takes *a person*, not how long the machine runs. MIT Technology Review called this the most misunderstood graph in AI, and that is the misunderstanding.
> Open the example — the ladder from "rename a variable" (2 min) to "port a component to another framework" (a day).
> Then note the axis multiplies: each step up is ten times bigger, so a straight line drawn here is already explosive growth. People still draw it too shallow.

> Three minutes in 2023. Eleven hours now. The shaded band is real uncertainty — that eleven hours runs from five to forty.

> **Test 4 · What it costs · the only one that falls**
> Hardest to draw, because it goes down. Hold the ability fixed at what GPT-4 could do in 2023 and ask what the cheapest model clearing that bar charges now. **208× cheaper in 23 months.**
> *“Every sentence that starts ‘it is too expensive to…’ has a shelf life, and the shelf life is months.”*
> Say out loud that the line stops in early 2025 because that is where the data stops.

> **Test 5 · The one that turns**
> **Open the puzzle first and let the room solve it.** Coloured squares fall to the bottom. People get it in about two seconds. That is the whole point — this is the test a child passes and a machine could not.
> Then: five years near zero, then 87.5% in one announcement. Let them draw it steeply. They have earned it.
> Then press **Show me**. Three months later, harder version, same model: 4%. Then March 2026, version three: the average human tester scored **48%**. GPT-5.6 Sol reached **38.3%** when allowed to retain reasoning between turns, versus **13.3%** with standard wiring. A **0.51%** bare-model result came from ARC Prize’s different game set and harness, so it is not a like-for-like fourth bar.
> **Pause here.** Do not talk over the bars. Wait for someone to say “wait, what?”

The scorecard

Reports how many they guessed low and the median factor they were out by. The line to land: the mistake is not pessimism. It is assuming there is one curve.

Act II — where all this came from (5 minutes)

Press **Play** and stop talking for thirty seconds. 123 releases arrive; the argument makes itself as the gaps between them shrink from months to weeks.

- **Hollow rings are models anyone can download and run.** Solid dots you can only rent. Say this once, clearly — most rooms have never heard the distinction.
- Filter to **Downloadable** and scrub. DeepSeek-R1 in January 2025 is where the shape changes.
- Filter by region. **China** is the surprise for most audiences — Alibaba, DeepSeek, Moonshot, Zhipu, Tencent, ByteDance, Xiaomi, and a food-delivery company. **Europe** has Mistral plus EU-funded efforts in Finland and across the union. The US has Meta, Google's Gemma, Microsoft's Phi, NVIDIA, IBM and a research institute that publishes its training data too.
- Hover any dot for what it was, its licence, and why it mattered.
- Worth saying plainly: “downloadable” is not “open source”. Several carry commercial restrictions. The field calls them open source anyway.

Act III — what it can actually do (5 minutes)

Six domains. Do not read them all — pick two or three for your room and use the jump buttons at the top.

- **Proteins** is the settled case, and it has a Nobel Prize. Good opener for a sceptical room.
- **Medicine** is the honest pairing: an AI-discovered drug in a real randomised trial — and, on actual patient records rather than exam questions, models performing 16–25 points *worse* than clinicians. Both peer-reviewed. Show both.

- **Weather** is the quiet one that touches everyone and is running in production in Europe.
- **Materials** is the cautionary one: a flagship "new" compound that was first reported in 1972 and was in the model's own training data.
- **Mathematics** is where the potential showed earliest, because a proof is checkable. Read Tao's quote off the screen: generation and checking are now fast, understanding is still human and still slow.
- **Software** holds the single most important number in the demo — see below.

> **The number to end on.** METR ran a randomised controlled trial: experienced developers, real tasks, codebases they knew well. With AI tools they were **19% slower** — and they had predicted a 24% speed-up beforehand, and still believed they had been 20% faster afterwards. Set that beside every productivity claim anyone in the room has heard. It is the strongest evidence in the demo that self-assessment about these tools is unreliable, in both directions.

Finish on **Cut from this demo** — seven claims researched and excluded, with reasons. If anyone challenges a number during the talk, that is the box that buys you credibility.

> **If someone asks whether you made the numbers up:** Settings → **Run the self-test**, live. It checks that every plotted point resolves to a declared source, that no source is cited then unused, that the hidden data genuinely postdates the shown data, that every result in Act III says what the machine did *and* what the humans did, and that the doubling time quoted in Act III matches the points plotted in Act I. Every source is listed and linked at the bottom of Act III.

The numbers, for answering questions

TEST	FROM	TO	SOURCE
Fixing real bugs	33.4% · Jun 2024	82.0% · Sep 2025	Anthropic, OpenAI model cards
PhD science questions	39% · Mar 2023	94.1% · 2026	OpenAI, DeepMind, Artificial Analysis
Working alone	3.5 min · Mar 2023	11.3 hr · Jun 2026	METR Time Horizon 1.1
Price at GPT-4 ability	$37.50/M · Mar 2023	$0.18/M · Feb 2025	Epoch AI
Grid puzzles, v1	~0% · 2020–2024	87.5% · Dec 2024	ARC Prize
Grid puzzles, v3	human average 48%	Sol 38.3% with memory; 13.3% standard	OpenAI / ARC Prize; bare 0.51% is a different run
Developers with AI tools	expected 24% faster	measured 19% slower	METR randomised trial

Questions you will get

“Aren't these tests rigged or leaked?”

Partly, and the demo says so in every caveat box. Concretely: the model that scored 87.5% on the grid puzzles had been trained on 75% of the public practice set, and that score cost about $4,560 per puzzle. Meta once submitted a tuned variant to a leaderboard that was not the model it released. One test's own answer key was found to have errors. All real. The 0.51% bare-model result is informative precisely because it is a different game set and harness, not a direct comparison with the 48%, 38.3%, and 13.3% public-game measurements — which is why the demo keeps the wiring and run conditions visible.

“So is it about to replace everyone or not?”

The honest answer this demo supports: the length of job it can finish is doubling roughly every three months for recent models; there is a whole category of task where standard wiring scored 13.3% and memory-preserving wiring 38.3% against a 48% human average; a separate bare-model run scored 0.51%; and in the best-controlled study, it made expert programmers slower while convincing them it had made them faster. All three are true, of the same systems, in the same months. Anyone confident in either direction is reading one curve and ignoring the others.

“Has it actually discovered anything, or just passed tests?”

Yes, and it is a short list, and it is worth being precise. A Nobel-recognised solution to protein structure. Weather forecasting now running operationally in Europe. Two published mathematical results in 2026 where the key idea came from a model and named mathematicians verified it. One drug through a randomised Phase 2a trial. Against that: a flagship “new material” that was first reported in 1972, and ten “solved open problems” that turned out to be ten literature citations.

“Why does the cost line stop in 2025?”

Because that is where Epoch stopped publishing. Drawing a dotted line to today would look like data and be a guess. Saying that out loud is the most persuasive thing in the talk.

Misconceptions to head off

- **“Test 3 shows how long the AI can run.”** It shows how long the *job* takes a person. Correct it every time; it will come back.
- **“Scores going up means it is generally getting smarter.”** Test 5 exists to break this. Same generation, different task and wiring: 94% on science questions, 13.3% standard or 38.3% memory-preserving on public ARC-AGI-3 games; the 0.51% bare-model result is a separate, non-comparable run.
- **“Open source AI is way behind / has already won.”** Four months behind on Epoch's index as of May 2026, and that predates Kimi K3. Also, “downloadable” is not “open source”.
- **“The curve will obviously keep going.”** The demo does not claim this and neither should you. It shows what happened. Four of the five extrapolations the room just made were wrong, which is a good reason for humility about the sixth.
- **“It obviously makes people more productive.”** The one randomised trial we have says the opposite for experienced developers on familiar code — and that they could not tell.

If you have twelve minutes instead of twenty-five

1. Opening screen, read the first line. 1 min.
2. Test 1 (real bugs) — establishes the game. 3 min.
3. Test 3 (working alone) — the strongest single chart. 3 min.
4. Test 5 (the puzzle) with the twist — the turn. 4 min.
5. Act III → Software → the 19%-slower trial. 1 min, and stop there.
